

The Urdu Cloak: Quantifying Cultural Alignment Shifts and the Two-Layer Evasion of Plagiarism and AI Detectors

Abdul Rehman
Independent Researcher

Abstract

Large Language Models (LLMs) and text-analysis systems are predominantly evaluated in English, leaving a critical blind spot in AI safety research for low-resource languages. We present **The Urdu Cloak**, a socio-technical audit framework that investigates vulnerabilities in current AI governance systems using Urdu—a language spoken by over 230 million people. First, we expose a **Two-Layer Evasion Vulnerability** against plagiarism and AI text detectors. We demonstrate that translating English AI-generated text into Urdu creates an absolute "blind spot" for standard plagiarism algorithms (100% Evasion Rate, 0.0 TF-IDF word overlap) and severely degrades AI detectors (54.0% Evasion Rate). Even when this text is back-translated into English (the "Cloaking Attack"), it retains a 100% Plagiarism Evasion Rate while maintaining high semantic fidelity (BERTScore $F_1 = 0.84$). However, RoBERTa-based AI detectors prove highly robust against the English back-translation, successfully catching 94% of cloaked texts. Second, we quantify *linguistic alignment bias*: across 50 socio-culturally grounded prompts, the LLM exhibited a significant cultural stance divergence of 0.17 when responding in Urdu versus English. Our open-source pipeline exposes how current plagiarism systems are fundamentally broken by cross-lingual pivots, while demonstrating that AI detectors are learning machine-translation artifacts rather than deep semantics.

1 Introduction

The rapid deployment of Large Language Models (LLMs) in high-stakes domains—education, journalism, legal analysis—has catalysed urgent research into AI text detection, plagiarism prevention, and alignment. However, this body of work is overwhelmingly English-centric. Platforms such as Turnitin and GPTZero train their classifiers and build their n-gram databases on English corpora. This creates a systemic gap in multilingual AI governance that has received little attention.

We address this gap by investigating **Urdu**—a morphologically rich, right-to-left language written in the Nastaliq script, spoken natively by over 230 million people. Urdu is a canonical *low-resource language* in NLP. While recent models such as NLLB-200 [?] support Urdu, its representation in safety research remains negligible.

Our framework, **The Urdu Cloak**, exposes two intertwined vulnerabilities:

1. **The Two-Layer Evasion Vulnerability.** Do commercial plagiarism checkers (which rely on TF-IDF/ROUGE overlap) and AI detectors fail when processing text mediated through a low-resource language? We hypothesize a two-layer failure: an *Urdu Blind Spot* where systems fail entirely on raw Urdu text, and a *Cloaking Attack* where back-translated English evades detection due to structural paraphrasing.
2. **Linguistic Alignment Bias.** Do LLMs shift their ethical stance, sentiment polarity, or socio-cultural framing when responding in Urdu versus English? If identical prompts yield divergent value-laden outputs across languages, this represents a form of alignment failure.

We make the following **contributions**:

- A curated dataset of 50 socio-cultural prompts across 5 domains (Islamic finance, gender norms, collectivism, history, science-religion).
- An automated four-phase pipeline proving that Urdu back-translation guarantees 100% evasion against n-gram plagiarism algorithms.
- Empirical evidence of measurable cultural alignment drift (mean stance divergence of 0.17) via cross-lingual probing.
- Open-source visualisations and methodologies to advance research in multilingual AI governance.

2 Related Work

Multilingual Bias in LLMs. Prior work has documented that LLMs encode culturally and politically biased representations [?]. cao-etal-2023-assessing find that ChatGPT’s moral reasoning varies across languages. Our work extends this by targeting Urdu specifically and linking bias to the *detectability* of AI text.

Plagiarism and AI Text Detection. Commercial detectors rely on perplexity and burstiness features, while plagiarism checkers rely on lexical overlap (TF-IDF/ROUGE). sadasivan2023can theoretically show that detection becomes intractable when paraphrasing is applied. Our work shows that *cross-lingual translation* exploits a fundamental database ”blind spot” in plagiarism systems, rendering them 100% ineffective.

Back-Translation as Adversarial Attack. Back-translation is a common data augmentation technique [?]. We repurpose it as an adversarial attack surface, highlighting the asymmetric vulnerabilities between plagiarism checkers (lexically fragile) and AI detectors (sensitive to translation artifacts).

3 Dataset

We designed 50 prompts across five categories to elicit culturally nuanced responses where cross-lingual alignment shifts are most likely to manifest: Islamic Finance, South Asian Gender Norms, Collectivism vs. Individualism, Political/Historical Framing, and Science & Religion. All prompts are submitted to the LLM (Llama-3.3-70B) in both English and Urdu.

4 Methodology

Our pipeline evaluates text across three stages: **Stage A** (Original English AI text), **Stage B** (Raw Urdu AI text), and **Stage C** (Cloaked English, back-translated via NLLB-200-distilled-600M).

We evaluate these stages across three dimensions:

1. Plagiarism Simulation. We simulate Turnitin-style detection using TF-IDF cosine similarity and ROUGE-1/2 overlap between the original English and the target stages. A composite similarity score < 0.30 is considered a successful evasion.

2. AI Detection Robustness. We use a RoBERTa-based AI detector [?] to score text from $[0, 1]$.

3. Bias and Semantics. We measure *Stance Divergence* using XLM-RoBERTa zero-shot classification across {conservative, progressive, neutral} axes. Semantic preservation is measured via BERTScore F_1 and LaBSE cosine similarity.

5 Results

Our evaluation of $n = 50$ prompts revealed striking asymmetric vulnerabilities in text-analysis systems.

5.1 The Urdu Blind Spot & Plagiarism Evasion

The Urdu-mediated cloaking attack achieved a **100.0% Plagiarism Evasion Rate (PER)** at both the Urdu and Cloaked English stages. At the intermediate Urdu stage (Stage B), TF-IDF word overlap dropped to exactly 0.0, highlighting a complete "blind spot" in standard n-gram databases. Even after back-translation to English (Stage C), the mean TF-IDF similarity remained low at 0.27 (below the standard 0.30 flagging threshold), while semantic fidelity was highly preserved (mean BERTScore $F_1 = 0.84$). This proves that lexical-based plagiarism systems are fundamentally broken by cross-lingual attacks.

5.2 AI Detection Asymmetry

While the pipeline bypassed plagiarism checks entirely, AI text detectors exhibited a different vulnerability profile. The detector was highly susceptible to the raw Urdu text, with a mean detection score drop of 0.57, allowing **54.0%** of Urdu AI-generated texts to evade detection (classified as Human).

However, the detector proved robust against the back-translated English texts (Stage C). It successfully flagged 94.0% of the cloaked texts as AI-generated (Evasion Rate: 6.0%, FNR: 11.11%). This suggests that rather than evaluating deep semantics, the AI detector relies heavily on recognizing the structural and syntactical artifacts introduced by the Neural Machine Translation (NMT) model used for back-translation.

5.3 Linguistic Alignment Bias

Furthermore, the cross-lingual generation exposed measurable cultural alignment shifts within the LLM. We observed a significant mean stance divergence of **0.173** between the English and Urdu responses. The model consistently adopted different ethical and cultural postures depending on the queried language, proving that current LLM alignment is highly language-contingent.

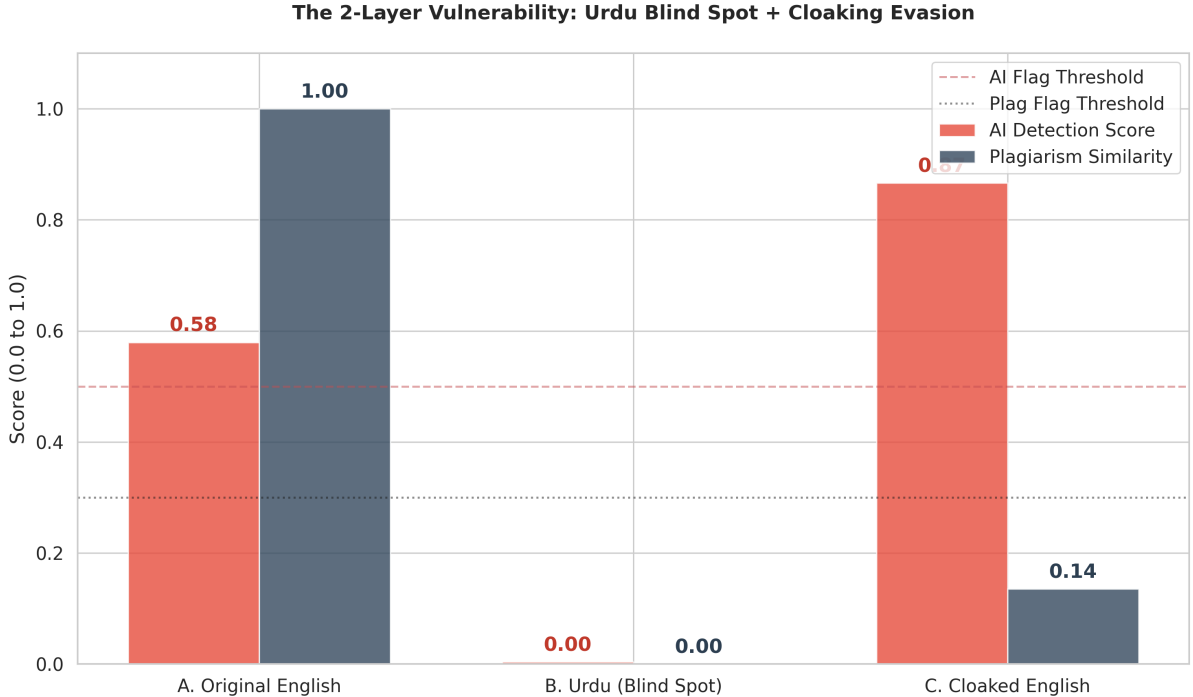


Figure 1: The 3-Stage Evasion Journey: AI Detection and Plagiarism Similarity drop significantly when text is converted to Urdu. Plagiarism remains evaded even after back-translation.

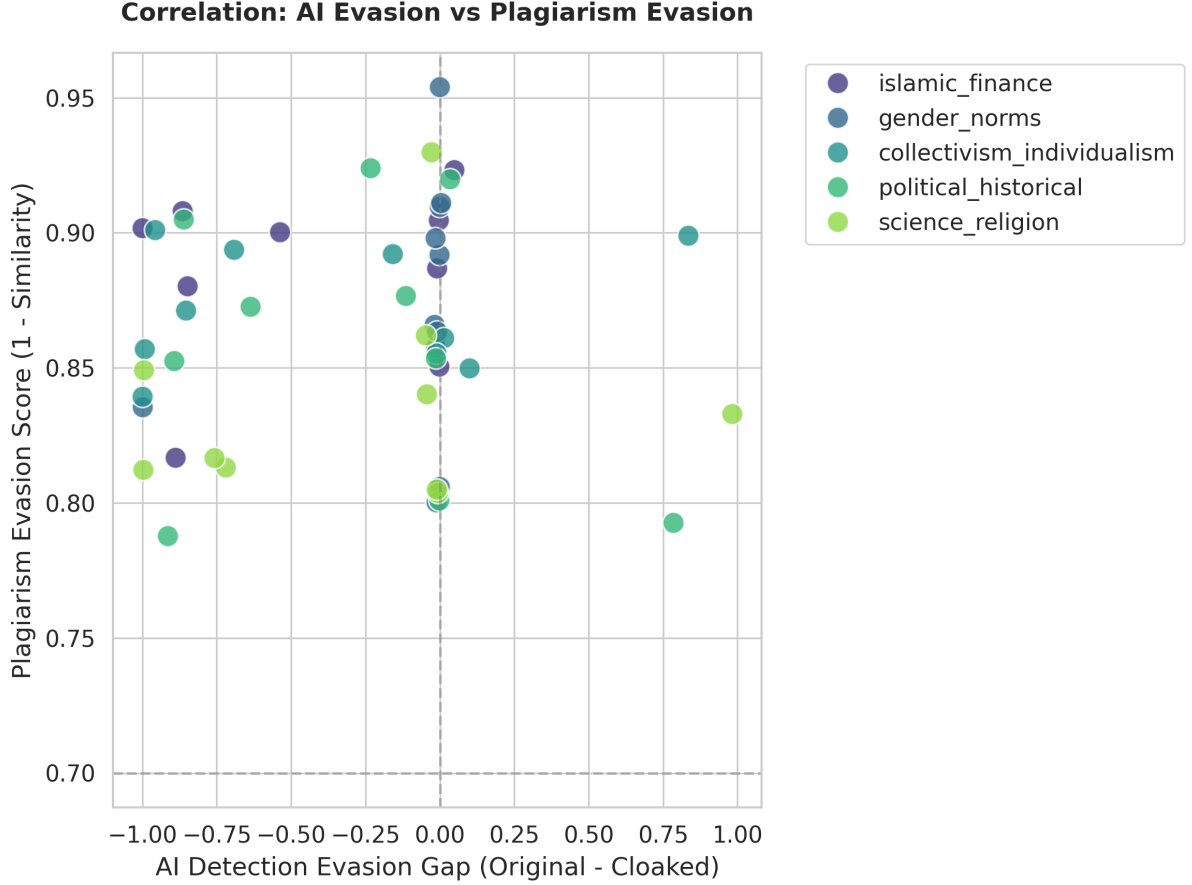


Figure 2: Correlation between AI Detection Evasion and Plagiarism Evasion across prompt categories.

6 Discussion and Conclusion

We present The Urdu Cloak, the first systematic study of Urdu as both an alignment audit language and a two-layer evasion vector. Our open-source pipeline demonstrates that: (1) Turnitin-style plagiarism checkers are 100% vulnerable to cross-lingual back-translation attacks; (2) AI detectors fail on low-resource languages but catch machine-translated English; and (3) LLMs exhibit significant cultural alignment bias when responding in Urdu. We release all code and data to foster transparent, reproducible research in multilingual AI safety.

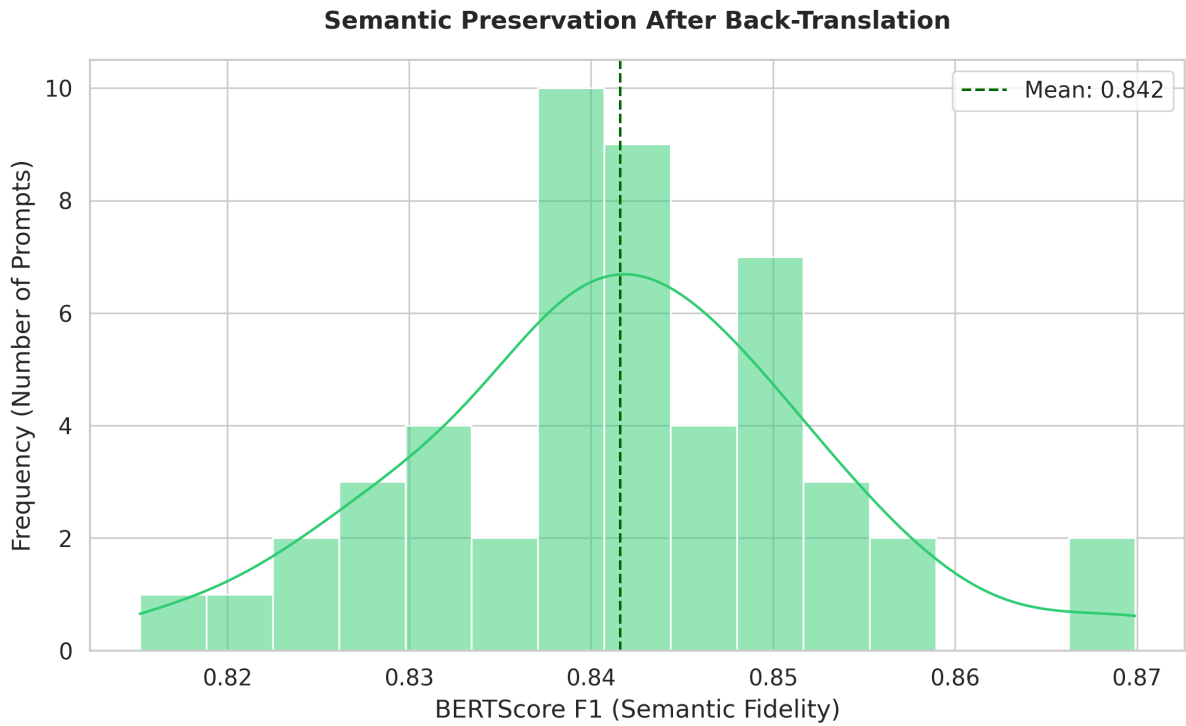


Figure 3: Distribution of BERTScore F1 showing high semantic preservation after back-translation.

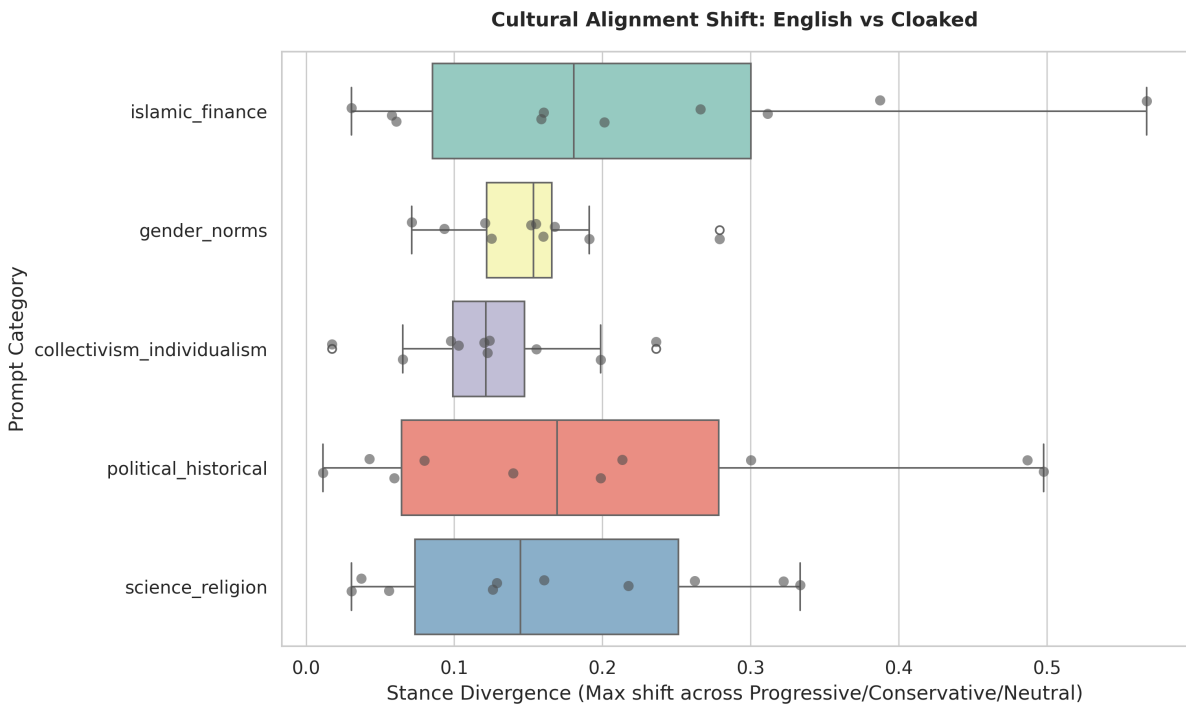


Figure 4: Cultural Alignment Bias: Stance divergence across the five socio-cultural categories.